



FEEDPROBE

Agent-Based Social Media Virality Prediction

Upload your product demo — AI personas representative of your target audience (and outside it) watch it, act on it, and an Explore-style algorithm pushes it to the people most like them.
Know how it performs before you post.



WATCH THE DEMO · click to play the screen recording

VIDEO: feedprobe.surge.sh/watch.html

SITE: feedprobe.surge.sh

CODE: github.com/aaryankj80-web/feedprobe

PREDICT BEFORE YOU POST

BETA REPORT

THE PROBLEM & WHO IT IS FOR

THE PROBLEM

Launching a product demo is a gamble — and the algorithm is a black box.



Blind launches

You post and wait. No feedback on the hook, the pacing or the CTA before real money is spent.



Black-box algorithms

Nobody can explain why one reel pops and an identical one dies. Signal stays invisible until it is gone.



Expensive iteration

Agencies and teams burn budget testing variants against live audiences — slow, public and costly.

#6Slow feedback

Days of analytics to learn what a five-second hook check could have told you up front.

TARGET USERS



Product Marketers

Validate hooks, angles and calls-to-action before launch day — with numbers, not hope.



Indie Founders

No ad budget to gamble. Test the demo once against personas instead of paying for impressions.



Social Media Managers

Post with confidence. Every upload ships with a predicted reach, engagement and virality read.



Agencies & Studios

De-risk client campaigns, show the prediction data, and justify creative spend to stakeholders.

WHY AN AGENTIC SOLUTION

WHY SIMULATION

- Virality is emergent: thousands of individual choices, not one decision.
- Static analytics say what happened. Simulation says what will happen.
- Cascades need many agents with many behaviours — not averages.

WHY AGENTS

- Personas carry demographics, interests and behaviour biases.
- Pools include your target demo AND people outside it.
- Agents act — watch, like, comment, share, skip — the signals algorithms feed on.

WHY NOW

- Deterministic and cheap: same seed, same result, in milliseconds.
- Test dozens of hooks, formats and audiences before launch.
- Agent platforms finally make whole-population tests practical.

TECHNICAL APPROACH & METHODOLOGY

1 · UPLOAD DEMO

 Drop the video; title, tags and length auto-extract. Analysis is metadata-based today, vision-ready tomorrow.

2 · BUILD PERSONAS

 200+ agents with demographics, interests and biases — target pool plus breakout pool.

3 · AGENTS ACT

 Each persona watches, likes, comments, shares or skips, driven by interest fit.

4 · EXPLORE PROPAGATION

 Engaged agents push the video to related agents, wave by wave — like Explore.

5 · PREDICT

 Reach, engagement rate, amplification, virality score and breakout %.

6 · INSIGHTS

 Rule-based tips: hook strength, shareability, best-fit clusters, next steps.

THE COMPLETE FLOW



SYSTEM ARCHITECTURE



Major components of the Feedprobe agentic simulation — every component feeds the central controller

PLANNING

Wave planner — seeds the audience & schedules
Explore steps

EXTERNAL SYSTEMS

Explore-style feed replica · vision / LLM brain
(plug-in)

RETRIEVAL

Content ingestion — metadata, tags, duration
extraction

TOOLS

Persona generator · content scoring · propagation
engine

AGENT / CONTROLLER

Orchestrates personas, waves & actions
routes every signal across the system

MEMORY / STATE

Agent pool, wave history & full action log

FAILURE HANDLING

Deterministic seeds · honest zero-engagement reporting

EVALUATION / VERIFICATION

Metrics & insights — reach, virality, rule-based checks

HUMAN INTERACTION

Upload, sliders, seed control, replay & the report

SOURCE CODE & REPOSITORY · github.com/aaryankj80-web/feedprobe (public)

Source in one self-contained HTML file · Setup: open it in any browser · Zero dependencies · No env config · README · Architecture docs



FEASIBILITY, VIABILITY & EXPECTED IMPACT

FEASIBILITY AND VIABILITY

FEASIBILITY

- Runs 100% in the browser — no server, no install.
- Deterministic per seed: same inputs, same prediction.
- 200+ personas simulated in milliseconds.
- Metadata-based today; vision/LLM brain ready to plug in.

POTENTIAL CHALLENGES & RISKS

- Frame-level video understanding still needs a vision model.
- Model must be calibrated against real platform data.
- Platform algorithms drift — tuning needs periodic refresh.

STRATEGIES / APPROACH

- Plug in Whisper / CLIP to analyze real video frames.
- Calibrate engagement curves with historical post data.
- A/B simulate hooks and audiences before every launch.

EXPECTED IMPACT

De-risk every launch

Predict reach and engagement before posting — not after.

Higher hit rate

Spend creative effort where the simulation says it works.

Faster iteration

Test dozens of variants in minutes, not days of live data.

Budget efficiency

Fewer failed campaigns, more confident media spend.